S2CIFTNet: Spatial–Spectral Coupling and Interactive Fusion Transformer Network for Hyperspectral Image Classification

Haixu Yang, Shuli Cheng, *Member, IEEE*, Anyu Du, and Jun Zhang

***Abstract*—Recently, the fusion network model using convolutional neural networks (CNNs) and Transformers has become a research hotspot in hyperspectral image classification (HSIC). However, these models applied to HSIC often adopt a single-branch heterogeneous architecture, which only achieves the fusion of the local details of CNNs and the global attention mechanism of Transformers through simple concatenation. Challenges remain to address both the nonlinear relationships between substantial redundant information in local complex features and the generalization ability at a single scale. To address these issues, we propose the spatial–spectral coupling and interactive fusion Transformer network (S2CIFTNet), which enables efficient extraction of multiscale spatial–spectral features, precise modeling of nonlinear feature relationships, and effective fusion of multimodal features. We have designed a spatial–spectral fusion second-order distance covariance (S2FDC) module, which can perform multiscale spatial–spectral feature extraction and effectively integrate different branch features through second-order distance covariance (SODC), accurately capturing the complex interdependency structures between variables, thus achieving cross-scale spatial–spectral information fusion. Additionally, we designed an interactive attention fusion (IAF) module that maps spatial–spectral features into feature markers and dynamically integrates these representations to extend the conventional multichannel self-attention mechanism. This extension promotes sufficient interaction and fusion between diverse input information, thereby achieving effective capture of both local details and global context, which significantly enhances the classification performance of the model. Experimental results show that on four public datasets, S2CIFTNet achieves excellent classification results with fewer training samples compared to existing methods. The source code link is available at https://github.com/huitailang99/S2CIFTNet**

***Index Terms*—Convolutional neural networks (CNNs), hyperspectral image classification (HSIC), multiscale spatial–spectral feature, Transformer.**

I. Introduction

IN THE field of remote sensing, hyperspectral image classification (HSIC) constitutes a critical task. There are many advanced technologies such as super-resolution (SR) technology [1], hyperspectral images (HSIs) technology, and multisource data fusion technology, to name a few. Among them, as a key technology in the field of remote sensing imaging, hyperspectral images technology captures the reflection spectral information of objects in tens to hundreds of continuous narrow bands, providing unique data advantages for fine classification of land features. As a result, HSI has been applied in multiple remote sensing fields, including environmental monitoring [2], [3], urban planning [4], [5], agricultural production [6], [7], and [8], and national defense [9], providing important technical support. Over the past few decades, researchers have come up with a variety of HSIC methods. According to the feature extraction modes, these methods generally fall into two categories: handcrafted methods and deep learning (DL)-based methods.

Traditional handcrafted methods include support vector machines [10], random fields [11], K-nearest neighbor [12], and principal component analysis [13]. However, these traditional methods rely on manual feature extraction, which limits their ability and ultimately results in lower classification accuracy.

Methods based on DL include convolutional neural networks (CNNs) [14] and Visual Transformers (ViTs), among others. Lee and Kwon [15] proposed a 2DCNN to jointly utilize the spatial and spectral features of HSI, but due to the deep fully convolutional structure, the number of parameters and computational complexity are relatively high. In addition, Hamida et al. [16] utilized 3DCNN to synchronously process spatial–spectral information. However, the 3DCNN operation increases the computational load of the model, resulting in a large number of parameters. To address this issue, Roy et al. [17] proposed HybridSN, which combines 3DCNN and 2DCNN to obtain spatial–spectral information and abstract-level spatial features. Although HybridSN achieves good classification results, it does not explicitly integrate attention mechanisms, resulting in poor flexibility in feature selection and reinforcement. The above CNN-based models are good at extracting local spatial–spectral information from HSI. However, they also exhibit complex structures, large parameter quantities, and significant computational costs. Its small sample generalization is weak, especially when training samples are scarce; the model is prone to overfitting, leading to degradation of classification performance.

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Although the algorithm model based on CNNs proposed by the aforementioned researchers has made significant progress in spatial–spectral feature extraction, its limitation of relying on single-scale modeling leads to insufficient utilization of multilevel spectral correlations and spatial contextual information. To address this issue, researchers have proposed a series of multiscale dual-branch network architectures. The network framework proposed in [18] for HSIC using a dual-branch dual attention network for spatial–spectral feature extraction has been validated as an effective HSI classification method. In order to obtain global modeling capability and extract more discriminative features, Zhang et al. [19] proposed a framework that combines multicriteria evaluation and dynamic removal strategy, which dynamically removes redundant bands by quantifying their spectral quality, spatial consistency, and classification contribution. Ghaderizadeh et al. [20] proposed a method, which innovatively embeds channel spatial attention modules in both spectral and spatial branches. Through an adaptive feature reweighting mechanism, it dynamically enhances discriminative feature expression and improves the discriminative ability of feature representation. Zhou et al. [21] proposed a method that integrates the multihop graph inference mechanism of GCN with the multiscale feature extraction capability of CNN and constructs a dual-stream heterogeneous architecture to synergistically model the spatial correlation characteristics of hyperspectral images. Meanwhile, Shu et al. [22] proposed a dual feature aggregation network that simultaneously aggregates spectral and spatial features and designs a feature fusion module to fully utilize the complementary information between the two. While the aforementioned dual-branch structure extracts rich multiscale spatial–spectral information, it significantly increases computational complexity and resource consumption.

In the case of texture similarity and continuous spectral bands, extracting discriminative features is a challenge. Therefore, identifying and utilizing the similarity between different spectral bands is crucial for feature extraction. Cheng et al. [23] combined covariance with Transformers, where the Transformer uses nonparametric covariance attention to capture the correlation between different bands, enabling the network to focus more on useful spectral bands. Fang et al. [24] introduced a novel local covariance matrix (CM) approach to capture the full correlation across different spectral bands. Zhao et al. [25] developed a density peak clustering-based CM method for hyperspectral image feature extraction. This technique not only enhances feature extraction but also minimizes intraclass variations and interclass interference. Cheng et al. [26] combined distance covariance with a cross-attention mechanism to capture nonlinear feature information. In the aforementioned studies on covariance, a common approach is to directly process the CM, neglecting the need to analyze the matrix from multiple perspectives. This method does not effectively simplify complex multidimensional data to highlight the most critical spectral features, thus simplifying the data processing process. Highlighting the differences between two covariance matrices and enhancing the representation of the model through nonlinear transformations can help capture subtle spectral variations, which are crucial for distinguishing between different feature classes.

In recent years, Transformer-based network models have a significant impact on the field of natural language processing (NLP). Dosovitskiy et al. [27] introduced a Visual Transformer (ViT), which was the first to apply the Transformer architecture to computer vision tasks. It decomposes the image into serialized image blocks and inputs them into the Transformer model for processing. Due to their powerful global modeling capabilities, researchers have developed a series of Transformers and successfully applied them to HSI classification tasks [28], [29]. For example, Hong et al. [30] proposed a Transformer-based network that captures local sequence information of adjacent bands through group spectral embedding and then uses cross-layer skip connections to input shallow information into the Transformer to reduce information loss. Gao et al. [31] proposed the spatial–spectral perception network for multisource remote sensing image classification. This network achieves effective collaborative classification of multisource data by extracting spatial relationships from high-resolution images and spectral relationships from spectral images. Sun et al. [32] proposed the memory-augmented spectral–spatial Transformer, which utilizes the Transformer's powerful processing of spectral sequence information. Jiang et al. [33] proposed graph generative structure-aware transformation by incorporating node embedding and position encoding as essential components in the transformation, in order to address the issue of position encoding losing physical meaning in classification tasks. Zhao et al. [34] solved the problem of parameter overload in hyperspectral data processing and proposed a new method based on lightweight ViT, namely, group separable convolutional ViT. Ding et al. [35] combined spectral–spatial convolution fusion with long-range dependency modeling, enabling adaptive capture of local features in hyperspectral images. Han et al. [36] combined subpixel-level spectral variability modeling with nonlinear mixed characteristics analysis, improving the ability to analyze mixed pixels while enhancing robustness to spectral variability. Although Transformer-based structures perform well in terms of performance, their self-attention mechanism can only learn features from input data within a single feature space. This method does not fully consider the interaction and fusion between features and is limited by its focus on individual features.

Based on the above analysis, we propose spatial–spectral coupling and interactive fusion Transformer network (S2CIFTNet) for HSIC. Specifically, we have designed a new spatial–spectral fusion second-order distance covariance (S2FDC) module. This module can deeply study the spatial and spectral details of images on branches of different scales and integrate the features of each branch through second-order distance covariance (SODC), effectively capturing the nonlinear relationships between variables and achieving the fusion of spatial–spectral information at different scales. In addition, we have designed an interactive attention fusion (IAF) module that includes feature markers T_n, T_m, and T_f, as well as cross fusion attention mechanism (CFAM) block.

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By dynamically integrating these feature representations mapped by spectral spatial features, we extend the traditional multichannel self-attention mechanism and enhance the efficiency of multimodal information flow and integration. This not only extracts rich global spectral information but also strengthens the ability to capture local spatial information, resulting in more accurate performance in HSIC. The main contributions of this work are summarized as follows.

1) We propose an S2CIFTNet. By integrating the advantages of CNN and Transformer, this architecture performs interactive fusion of multiscale spatial–spectral information and captures the band relationships across different scales through SODC, addressing the issue of weak generalization in small sample scenarios.
2) To address the limitations of existing multiscale spatial–spectral feature extraction techniques, we propose a S2FDC module. This module cleverly applies deep convolution with convolution kernels of varying sizes across multiple spectral channels, effectively extracting multiscale spatial–spectral features that encompass both local details and global structures. Additionally, we employ SODC to explore the nonlinear relationships between different spectral bands, thereby providing the model with a deeper understanding of spectral features.
3) In response to the complexity of feature dependencies in hyperspectral image processing and the limitations of existing attention mechanisms in capturing broad dependencies, we propose an IAF module. This module provides rich feature representations for the subsequent interactive attention mechanism through multiscale mapping and pooling operations, effectively addressing the issue of insufficient feature representation. It enables the model to capture a wider range of feature dependencies, thereby enhancing the model's representational capability and performance in handling complex scenarios.

We have conducted comprehensive experiments on four benchmark HSI classification datasets: Indian Pines, Salinas, Botswana, and WHU-Hi-LongKou. The experimental results demonstrate that our method achieves state-of-the-art performance, outperforming existing HSI classification techniques.

The structure of this article is organized as follows. Section II reviews the related work in the field of HSI classification. Section III elaborates on the method proposed by us. Section IV presents the experimental results and provides an in-depth analysis. Finally, Section V summarizes the research work and offers prospects for future research directions.

II. Related Word

HSIC is of great significance in the field of remote sensing. More and more researchers are dedicated to HSI classification tasks and have developed a series of network models to improve classification performance. This section is divided into three parts to briefly summarize relevant work, namely, CNN-based HSI classification method, Visual Transformer-based HSI classification method, and CNN combined with Visual Transformer-based HSI classification method.

A. CNN-Based HSI Classification Methods

CNN has been widely used in HSI classification due to its excellent characteristics, such as local perception, parameter sharing, and translation invariance, and has achieved good classification results. Hu et al. [37] proposed a simple and effective 1DCNN network with a five-layer structure, each layer being implemented on spectral features, which can effectively extract spectral features. However, 1DCNN ignores the rich spatial information in HSI. Therefore, Cai et al. [38] proposed a novel 2DCNN neural network that uses different convolution kernels to extract information from different spatial positions and utilizes penalty and reward strategies to enhance features. However, when considering the spectral and spatial information independently in hyperspectral images, the classification performance is not satisfactory, and the rich 3-D structural features in hyperspectral images are not fully utilized, while the joint features of spatial and spectral information are ignored. To address these issues, Hamida et al. [16] utilized the 3DCNN to synchronously process spatial–spectral information and used 3DCNN to extract spatial–spectral information from HSI for joint modeling, significantly improving the accuracy of classification. However, 3DCNN has higher computational complexity compared to 2DCNN. Second, the convolution operation of 3DCNN may not be flexible enough for extracting complex spectral information from hyperspectral data, as it mainly focuses on spatiotemporal relationships, which may lead to insufficient expression ability of the model on spectral features. Therefore, Roy et al. [17] proposed HybridSN, which combines 3DCNN and 2DCNN to obtain spatial–spectral information and abstract-level spatial features. For multivisual processing of HS image classification tasks, Liu et al. [39] proposed a heterogeneous dual enhanced network based on hybrid CNNs joint multiscale dynamic GCNs (HDECGCN) that extracts complementary spatial–spectral information by enhancing CNN and GCN subnetworks, thereby improving the classification accuracy. In addition, Liu et al. [40] proposed a dual-branch encoder–decoder structure that combines spatial and spectral enhancement branches and achieves complementary enhancement of spatial spectral information through a multiscale feature fusion module. Although CNN-based networks have good performance in feature extraction, it is difficult to fully utilize spatial–spectral information by relying solely on single-scale feature extraction. Ding et al. [41] propose an unsupervised framework that captures both the self-expressive subspace characteristics and local structural information of data and enhances feature representation capability through graph convolution embedding. Although the CNN-based networks mentioned above have good performance in feature extraction, relying solely on a single scale to extract features makes it difficult to fully utilize spectral–spatial information. In addition, due to inherent architectural flaws, CNN can only extract shallow local spatial–spectral information and cannot capture long-range dependencies. Therefore, some researchers have

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begun to focus on HSI classification tasks based on Visual Transformers.

B. Visual Transformer-Based HSI Classification Method

In recent years, with the outstanding performance of Transformer in NLP and CV fields, researchers have begun to introduce it into HSIC tasks to explore its potential in spatial–spectral information modeling and HSI classification. Unlike CNN, which uses static filters to extract local information, VIT dynamically captures long-range dependencies and remote interactions in multihead self-attention (MHSA) through parallel computing. Li et al. [42] proposed a pulse neural network that combines an adaptive feature self-attention mechanism to enhance feature expression ability through the self-attention mechanism and utilizes the spatiotemporal dynamic characteristics of pulse neurons to improve classification performance. Ma et al. [43] proposed a self-supervised learning strategy to randomly mask hyperspectral image blocks and reconstruct masked regions using an encoder–decoder structure. Qin et al. [44] proposed a framework that combines a language enhancement module with a two-layer contrastive learning network, which uses natural language description to assist in spectral spatial feature learning and constructs contrastive loss functions at the pixel and category levels to enhance the model's discriminative ability for unknown categories. Wang et al. [45] proposed a weak–strong graph contrastive learning network, which constructs weak and strong enhanced graphs and utilizes contrastive learning strategies to maximize mutual information between the two, thereby learning more discriminative spectral spatial joint feature representations. Liu et al. [46] proposed a multilevel prototype alignment framework, which constructs multilevel prototype representations of the source domain and target domain and uses alignment strategies to reduce interdomain differences. Although the performance of the Transformer methods mentioned above is satisfactory, they still fall short in capturing local features that are crucial for HSI classification. Therefore, relying solely on Transformers makes it difficult to handle complex HSI classification tasks.

C. CNN Combined With Visual Transformer-Based HSI Classification Method

The combination of CNN and Transformer can leverage their respective strengths, and this hybrid approach is highly sought after by many researchers. In order to fully tap into their complementary advantages, Zhao et al. [34] proposed a group separable convolutional Visual Transformer (GSC-ViT), which adopts a group separable convolution (GSC) module and a group separable multihead self-attention (GSSA) module, effectively reducing the number of parameters and improving the ability to extract local and global features. Ding et al. [47] propose an unsupervised framework that captures both structural homogeneity and local structural information of data and enhances the robustness of graph structure through adaptive filtering. Yang et al. [48] proposed an adaptive coupling Transformer network that combines the advantages of CNNs and Transformers to design an adaptive coupling module that dynamically fuses local features with global contextual information. Zhou et al. [49] also proposed a dual-branch convolutional Transformer network called DCTN. It uses 3-D convolution and 2-D group convolution to extract rich spatial–spectral information in parallel and fuses this information through an efficient interactive self-attention (ELSA) mechanism that performs in height, width, and spectral dimensions. Mohamed et al. [50] proposed a factorized Transformer architecture called FactoFormer. It utilizes a factorized architecture to enable the model to learn independently and fuse in both spectral and spatial dimensions, thereby more accurately capturing the complexity in hyperspectral data. Then, it is pretrained using masked image modeling (MIM) technology, which greatly improves the model's generalization ability on limited annotated data. Shi et al. [51] proposed MHCFormer, which combines CNN, Transformer, and Fourier transform for HSI classification. Capture complex relationships between spectra and space through multiscale spectral–spatial labeling and hierarchical labeling learning. Xi et al. [52] proposed MCTGCL, which combines the advantages of CNNs and Transformers and introduces a graph contrastive learning mechanism to enhance feature representation capability. Ding et al. [35] proposed a Transformer architecture for modeling long-range dependencies, called LRDTN, which combines the advantages of CNNs and Transformers, achieving deep fusion of spectral–spatial features and effective modeling of long-range dependencies. It can be seen that the joint modeling method of CNN and Transformer is highly sought after by people. However, it is still challenging to fully explore the multiscale spatial–spectral information in CNN and the local spatial information in Transformer.

III. Proposed Method

This section provides a detailed exposition of our proposed S2CIFTNet model. First, we elucidate the overall framework of the S2CIFTNet model and its workflow for HSIC. Second, we delve into the motivation of several key components in the S2CIFTNet architecture.

A. S2CIFTNet Framework

The overall architecture of our proposed S2CIFTNet framework is shown in Fig. 1. The model consists of four parts: the S2FDC, the IAF, a global average pooling (GAP) layer, and a SoftMax classifier incorporating a fully connected (FC) layer. This framework achieves fine-grained classification of hyperspectral remote sensing data through multimodal feature fusion and hierarchical feature modeling strategies.

This article adopts the spatial–spectral collaborative representation SODC method to enhance the correlation between spatial and spectral information, effectively extracting local multiscale spatial–spectral information, and simplifying the channel features of each part, enhancing the correlation of each part's channel. Then, a cross-scale feature fusion framework is adopted to reduce the number of channels in feature mapping, and different levels of features are modeled separately; the output of each level is the sum of features at different scales.

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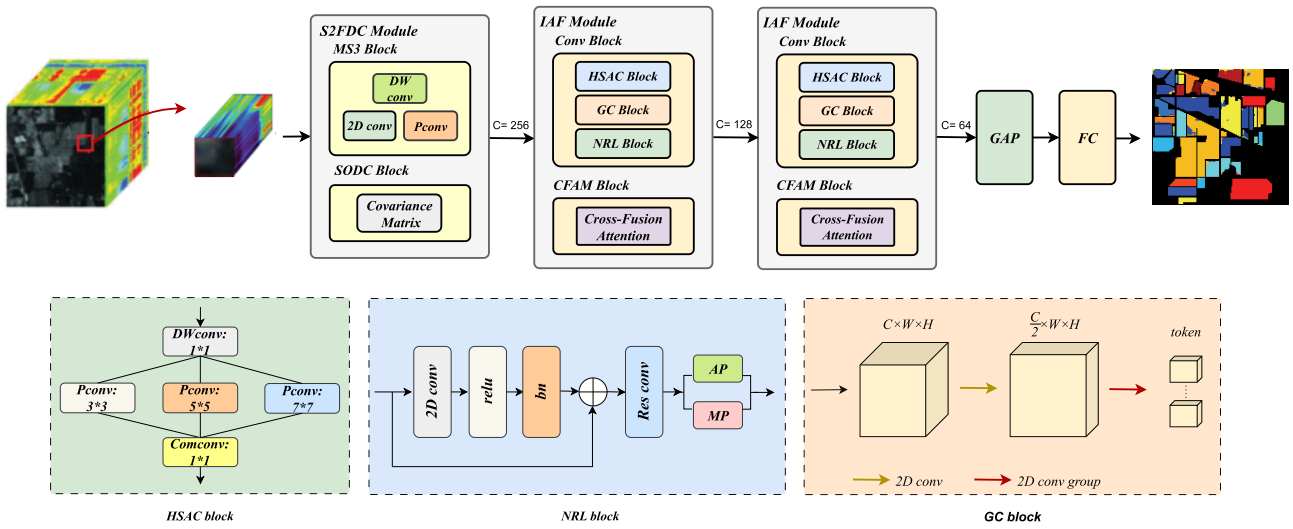


Fig. 1. Proposed S2CIFTNet network architecture. The network architecture was designed with two main modules: S2FDC and IAF. DWconv refers to the depthwise separable convolution, Pconv refers to the Partial Conv, Res conv stands for the composite residual convolution, MP refers to the max pooling, AP refers to the average pooling, GAP block denotes the global average pooling layer, and FC refers to the fully connected layer.

Each level consists of several different conv modules and a Transformer module, where conv modules are used to capture shallow spatial–spectral patterns of different granularities, and the extracted information is then transmitted to the Transformer module to establish global spectral dependencies and local spatial correlations. Finally, in the GAP layer, the output feature map is simplified to a 1×1 spatial dimension and then converted into a 1-D vector and input into a classifier composed of FC layers to obtain the final classification result.

B. S2FDC Module

Most HSIs use a single mode and scale to extract features, which is difficult to make full use of spectral spatial information and cannot integrate local details and global information to make the model more accurate, thus limiting the improvement space of classification performance. In order to characterize linear and nonlinear interactions in the spectral domain, as illustrated in Fig. 2, this article proposes an S2FDC module consisting of a multiscale spatial–spectral separation (MS3) block and an SODC block. The MS3 block is responsible for extracting multiscale spatial–spectral features from four different spectral intervals and then extracting and analyzing different spectral interval features pairwise through SODC, thereby significantly enhancing the correlation between spatial and spectral information and effectively extracting local multiscale spatial–spectral information. In a few sample classification tasks, it demonstrates the powerful ability of SODC. Finally, we concatenated the outputs of four different spectral bands into a feature output, processed the residuals, and embedded them into the IAF encoder module to further improve classification performance.

1) MS3 Block: In order to effectively extract local semantic information from HSI, we designed an MS3 block module consisting of a multiscale convolution composed of DWconv and Pconv, which extracts features from two different perspectives: spatial and spectral. First, use Pconv convolution to extract the spatial position features of each channel and allocate channels. The output of the Pconv convolution is expressed in the following equation:

$$X' = \text{Pconv}_{1\times1} C\,(X) \tag{1}$$

where X is a given input HSI feature map with a size of $X \in \mathbb{R}^{B\times C\times P\times P}$, P is the height and width of the feature map, C is the number of channels, B is the batch size, and the output HSI feature map is $X' \in \mathbb{R}^{B\times C\times P\times P}$. Then, according to a certain proportion, it is divided into four different spectral band intervals along the channel size, with each channel having a number of Z, to obtain four feature maps $X_Q \in \mathbb{R}^{B\times Z\times P\times P}$, $Q \in \{1, 2, 3, 4\}$. The specific formula is as follows:

$$\begin{aligned} X_1, X_2, X_3, X_4 &= \text{ChannelSplit}\,(X) \\ &= X'^{:,:Z}, X'^{:,Z:2Z}, X^{:,2Z:3Z}, X^{:,3Z:} \end{aligned} \tag{2}$$

where Z represents the number of channels in each spectral band interval. We determine the number of channels in each spectral band interval by $Z = r_c C$, where r_c is a set ratio (r_c = 0.25). X_1, X_2, X_3, and X_4 are the four different spectral band intervals after spectral segmentation.

Finally, in each different spectral band interval, we use DWConv spatial convolution blocks of different scales to extract the spatial position features of each channel. The specific operation is as follows:

$$\begin{aligned} X'_1 &= \text{DWconv}_{k_1\times k_1}\,(X_1) \\ X'_2 &= \text{DWconv}_{k_2\times k_2}\,(X_2) \\ X'_3 &= \text{DWconv}_{k_3\times k_3}\,(X_3) \\ X'_4 &= \text{DWconv}_{k_4\times k_4}\,(X_4) \end{aligned} \tag{3}$$

where X'_1, X'_2, X'_3, and X'_4 are the output HSI feature maps, with a size of $X_Q \in \mathbb{R}^{B\times Z\times P\times P}$, $Q \in \{1, 2, 3, 4\}$; k_1 represents the size of the rectangular kernel, set to 3 in this case; and k_2, k_3, and k_4 represent larger kernels, 5, 7, and 9, respectively. In this setting, we ensure that a portion of the input X'

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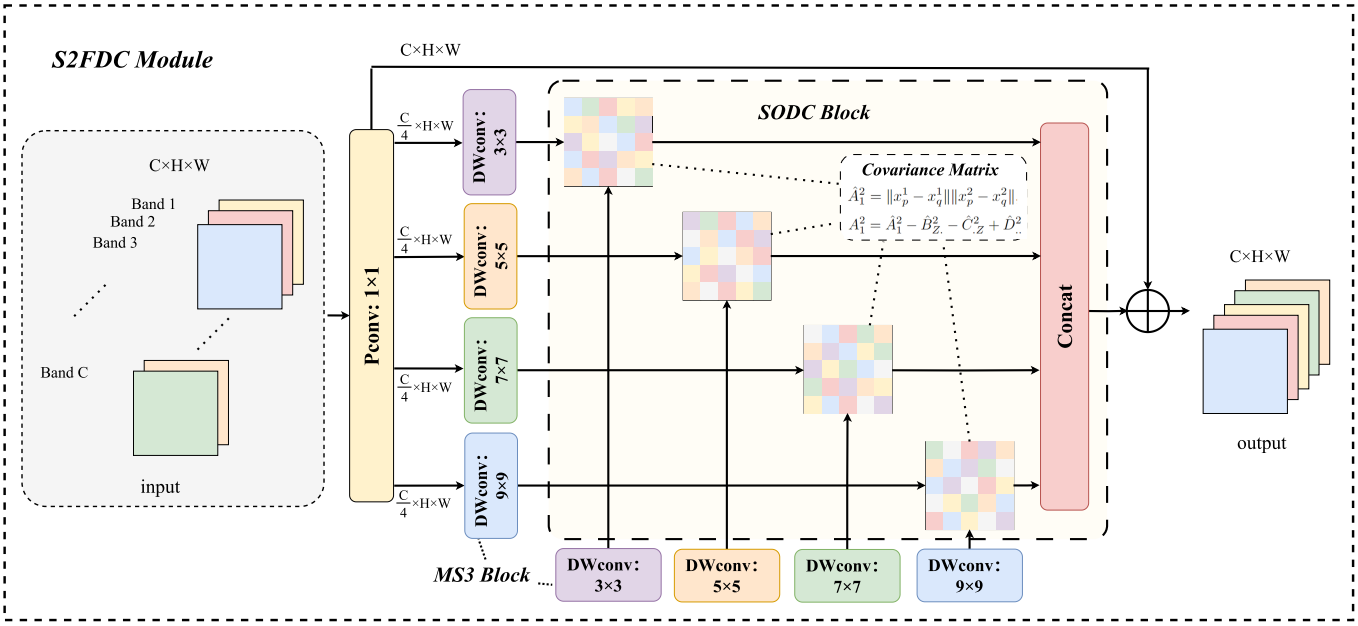


Fig. 2. Diagram of the S2FDC module structure, consists of two sub-blocks: MS3 block and SODC block.

is directly passed to the next layer, which helps maintain the integrity of the information and may reduce the risk of overfitting.

2) SODC Block: For second-order distance covariance module, in order to measure the nonlinear dependency relationship between two sets of variables, gain more advantages in processing high-dimensional data or nonlinear relationships, effectively handle feature selection or feature fusion, and solve the limitations of traditional covariance, we dynamically calculate fusion weights through SODC convolution in different spectral spaces and screen or recombine features to retain the nonlinear combination that contributes the most to classification. Moreover, feature selection was performed, and the SODC between each band and the target category was calculated to select a subset of bands with strong discriminability. In order to achieve a more comprehensive spectral spatial representation, the input consists of two sets of output sample features from the previous text, denoted as $X'_Q \in \mathbb{R}^{B\times Z\times P\times P}$, $Q \in \{1,2,3,4\}$, and $X'_{Q'} \in \mathbb{R}^{B\times Z\times P\times P}$, $Q' \in \{1,2,3,4\}$. The Euclidean distance between any two frequency bands in X'_Q and $X'_{Q'}$ can be expressed as follows:

$$\hat{A}_1^2 = \left\|x_p^1 - x_q^1\right\| \left\|x_p^2 - x_q^2\right\|, \quad p,q = 1,\ldots,Z \tag{4}$$

where x_p^1, $x_p^2 \in \mathbb{R}^{h^2}$ and x_q^1, $x_q^2 \in \mathbb{R}^{h^2}$, respectively, represent the pth and qth bands in X'_Q and $X'_{Q'}$. $\hat{A}_1^2$ is the Euclidean distance between x_p and x_q. This can be further expressed as the Euclidean distance matrix $\hat{A}_1^2 \in h_Z^+$. Finally, the distance CM A_1^2 can be expressed as follows:

$$A_1^2 = \hat{A}_1^2 - \hat{B}_{Z.}^2 - \hat{C}_{.Z}^2 + \hat{D}_{..}^2 \tag{5}$$

where $\hat{B}_{Z.}^1$, $\hat{C}_{.Z}^1$, and $\hat{D}^1$ represent the row mean, column mean, and overall mean of all elements in A_1^2, respectively. The same reasoning also leads to A_2^2, A_3^2, and A_4^2.

Next, in order to reveal the subtle changes in the statistical properties of different feature maps in HSI, we use a hybrid activation function (HAF) [26] to emphasize these differences. The final features X_Z^1, X_Z^2, X_Z^3, and X_Z^4 represent not only the original pixel information but also the additional discriminative ability brought by the average distance cross covariance difference, providing more accurate feature inputs for subsequent tasks. The subsequent equations are shown as follows:

$$\mathrm{HAF}(x) = \left(1 - \tanh(x)^2\right) \cdot \left(\mathrm{sigmoid}(x)\left(1 - \mathrm{sigmoid}(x)\right)\right) \tag{6}$$

$$Y = e^t \sqrt{\mathrm{HAF}\left(A_i^2\right)} \tag{7}$$

$$X_Z^i = -\left[Y - \frac{1}{C}(YI + IY) + \frac{1}{C^2} IYI\right] X'_Q + X'_{Q'} \tag{8}$$

$$X_{S2FDC} = \mathrm{Concat}\left(X_Z^1, X_Z^2, X_Z^3, X_Z^4\right) + X' \tag{9}$$

where C is the number of channels; I is the identity matrix; $X_{S2FDC} \in \mathbb{R}^{B\times C\times P\times P}$; X_Z^1, X_Z^2, X_Z^3, and $X_Z^4 \in \mathbb{R}^{B\times Z\times P\times P}$, $i \in \{1,2,3,4\}$, in the intraband covariance; and Q' is the paired spectral range ($Q' = Q$). t is the temperature parameter. By adjusting the smoothness of the function through HAF, it directly affects feature selection, decision boundaries, and model robustness. It can be dynamically adjusted based on specific task requirements (such as accuracy, speed, and scene complexity) and data characteristics, can help maintain information integrity, and may reduce the risk of overfitting.

C. IAF Module

The core architecture of the S2CIFTNet model aims to achieve local–global collaborative representation learning of input feature tokens. As illustrated in Fig. 3, the IAF module mainly consists of a gate control (GC) block, a heterogeneous scale-aware convolution (HSAC) block, a normalized residual learning (NRL) block, and a CFAM block. First, the channel dimension of feature mapping is dynamically adjusted through

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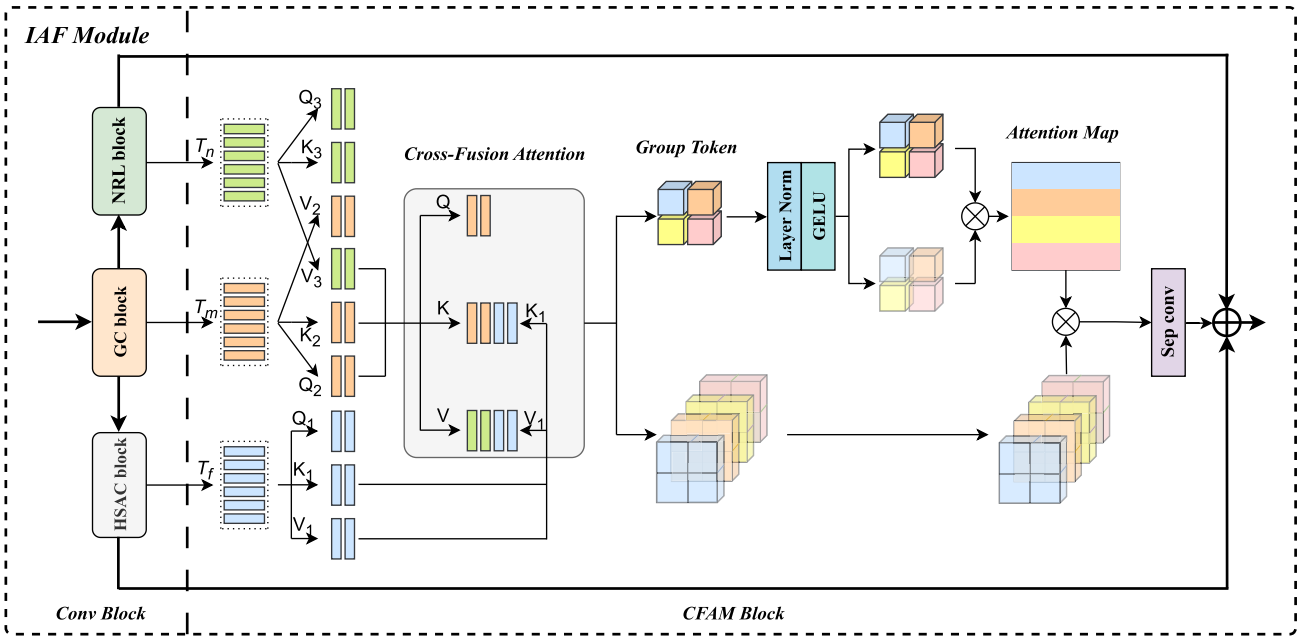


Fig. 3. Diagram of the IAF module structure, which consists of two sub-blocks: conv block and CFAM block.

the convolution structure of GC. Then, HSAC adopts multiscale fusion convolution to promote the deep fusion of local detail features and global semantic information. At the same time, the NRL module implements nonlinear feature enhancement on convolutional features through the collaborative effect of residual connections and multiscale pooling, significantly improving the robustness and discriminative power of feature expression. In the information integration stage, CFAM block performs Q, K, and V mapping on multisource features from T_n, T_m, and T_f, effectively improving the efficiency of multimodal information fusion.

1) GC Block: In order to reduce computational and parameter complexity, we propose a gated convolution structure that utilizes hierarchical channel modeling and feature abstraction mechanisms to achieve progressive representation learning of spatial–spectral features in hyperspectral images. For the output feature map $X_{S2FDC} \in \mathbb{R}^{B\times C\times P\times P}$ of the S2FDC module, two Conv2d grouped convolutions with different kernel sizes are used for feature abstraction compression, namely, 1×1 and 3×3 convolution kernels, forming a compact feature representation. The output feature map size is $X_{out} \in \mathbb{R}^{B\times C_{out}\times P\times P}$, $C_{out} = 0.5C$. Effectively extract advanced semantic features while reducing computational and parameter complexity, balancing performance and resource consumption.

2) HSAC Block: In order to further extract local details of HSI into global semantic information, we propose a multiscale model composed of depthwise separable convolution (DWconv) and partial convolution (Pconv). Specifically, the input feature map $X_{out} \in \mathbb{R}^{B\times C_{out}\times P\times P}$ is first processed through a DWconv, and then, the input feature map is processed through three different sized convolution kernels Pconv1, Pconv2, and Pconv3. Perform convolution operations on 3×3, 5×5, and 7×7 convolutions to extract local features at different scales. The specific formula is as follows:

$$\begin{aligned} X'_{out} &= \mathrm{DWconv}_{1\times 1}(X_1) \in \mathbb{R}^{B\times C_{out}\times P\times P} \\ X^1_{out} &= \mathrm{Pconv}_{k_m\times k_m}(X_2) \in \mathbb{R}^{B\times C_{out}\times P\times P} \\ X^2_{out} &= \mathrm{Pconv}_{k_n\times k_n}(X_3) \in \mathbb{R}^{B\times C_{out}\times P\times P} \\ X^3_{out} &= \mathrm{Pconv}_{k_r\times k_r}(X_4) \in \mathbb{R}^{B\times C_{out}\times P\times P} \end{aligned} \tag{10}$$

where C_{out} is the number of channels, and k_m, k_n, and k_r correspond to the convolution operations of 3×3, 5×5, and 7×7 convolution kernels, respectively, used to extract local features at different scales. Then, these outputs are connected along the channel dimension to form the final feature map X. Finally, the concatenated multiscale features are fused through a combine convolution to output a feature map with rich multiscale information. The specific equation is as follows:

$$X_{\mathrm{all}} = \mathrm{Concat}(X_{\mathrm{out1}}, X_{\mathrm{out2}}, X_{\mathrm{out3}}) \tag{11}$$

$$X_{\mathrm{HSAC}} = \mathrm{Comconv}(X_{\mathrm{all}}) \in \mathbb{R}^{B\times C_{\mathrm{out}}\times P\times P} \tag{12}$$

where $X_{HSAC} \in \mathbb{R}^{B\times C_{\mathrm{out}}\times P\times P}$ is the final feature map, which can simultaneously capture local details and global contextual information of the input feature map, thereby enhancing the model's perception ability of multiscale targets. Not only does it improve the expression ability of features, but it also makes the model more robust in handling tasks with large-scale changes.

3) NRL Block: In order to enhance gradient fluidity and avoid the degradation problem of deep networks, we propose a residual context aggregation module based on batch normalization enhancement. Specifically, first, feature recalibration is performed. The feature tensor $X_{out} \in \mathbb{R}^{B\times C_{\mathrm{out}}\times P\times P}$ input by GC is first compressed in the channel dimension through 2-D convolution to generate the basic feature representation X_{2-D}. The specific formula is as follows:

$$X_{\text{2-D}} = \mathrm{2Dconv}(X_{\mathrm{out}}) \in \mathbb{R}^{B\times C_{\mathrm{out}}\times P\times P}. \tag{13}$$

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Next, ReLU activation and batch normalization operations are sequentially applied to the output feature map $X_{2\text{-D}} \in \mathbb{R}^{B\times C_{\text{out}}\times P\times P}$ to obtain normalized features, effectively alleviating internal covariate shift and enhancing the nonlinear feature tensor. Then, connect the features with the original features to achieve residual connections and perform deep processing on the features through Resconv. The specific formula is as follows:

$$X_{\text{Res}} = \text{Resconv}\,(X_{\text{out}}) \in \mathbb{R}^{B\times C_{\text{out}}\times P\times P} \tag{14}$$

where Resconv is composed of two 2-D group convolutions and two 2Dconv. Finally, the recombined features are subjected to max pooling and average pooling, respectively, extracting the global max and average features. The results of max pooling and average pooling are concatenated in the channel dimension to form global contextual information. Improved the model's ability to understand the overall structure.

4) CFAM Block: To understand global contextual information and address the issue that local convolution operations may not fully capture global semantics. We propose a variant of the multihead attention mechanism, where the original memory information T_m combines additional information T_n and T_f. We repeatedly expand a fixed group token $M \in \mathbb{R}^{G\times C\times 1}$ to match the batch size of input T_m, T_n, and T_f, and concatenate it with three feature markers $T_i \in \mathbb{R}^{G\times C\times U}$, $i \in \{1,2,3\}$ in the last dimension to form a new feature marker $T_i^1 \in \mathbb{R}^{G\times C\times(U+1)}$, $i \in \{1,2,3\}$.

Next, use the weight matrices W_q, W_k, and W_v of three linear transformations to linearly map the new feature markers $X_i^1 \in \mathbb{R}^{G\times C\times(U+1)}$, $i \in \{1,2,3\}$ to Q, K, and V in sequence. The specific formula is shown as follows (17):

$$\begin{cases} Q_i &= X_i' W_q \\ K_i &= X_i' W_k \\ V_i &= X_i' W_v \end{cases} \qquad i \in \{1,2,3\} \tag{15}$$

where $Q_i, K_i, V_i \in \mathbb{R}^{G\times C_s\times(U+1)}$, $i \in \{1,2,3\}$; $C_s = 1/8C$, in which C is the number of channels; the patch sizes before grouping are $P \times P$ (12×12); G is the size of the grouped space, which is 9 (3×3); U is the number of groups, which is 16 (4×4); i is the category of the feature marker; and d is the dimension of the embedding vector.

To enhance the global visual representation ability, we will use a reshaping operation to adjust the dimensions of Q_i, K_i, V_i, from $G \times C_s \times (U+1)$ to $G \times 1 \times (U+1) \times C_s$. The resulting new key matrices $Q_i, K_i, V_i \in \mathbb{R}^{G\times 1\times(U+1)\times C_s}$, where $i \in \{1,2,3\}$. In this process, the key matrices and value matrices are dynamically concatenated, allowing the model to better utilize the feature information from the input data, thereby improving the classification model's understanding and prediction performance for hyperspectral data. This operation helps the model better distinguish features from different bands and enhances its understanding of the relationships between different wavelengths, contributing to more accurate classification of different categories within the hyperspectral data. The specific formula is shown as [see (18)]

$$\begin{cases} Q &= Q_2 \times s \\ K &= \text{Concat}\,(K_2, K_1) \\ V &= \text{Concat}\,(V_3, V_1) \end{cases} \tag{16}$$

where $s = D^{-0.25}$, D is the dimension of each attention head, and s is the reciprocal of the scaling factor. The key and value matrices are $K, V \in \mathbb{R}^{G\times 1\times 2(U+1)\times C_s}$, while the query matrix is $Q \in \mathbb{R}^{G\times 1\times(U+1)\times C_s}$. K_1, K_2, V_1, and V_3 are learnable parameters dynamically learned and processed through a series of operations to generate the new key matrix. To address the issue of mismatched dimensions between Q and K, we use the Einstein summation convention (einsum) to compute the Dot product between the Q and the K. The Dot product result, referred to as Dots, represents the similarity between each Q and K. This generates an attention score matrix, which is used to measure the degree of attention a query has to a key. This process helps to dynamically aggregate global information, enabling the model to capture long-range dependencies. The specific formula is shown as [see (19)]

$$\text{Dots} = \frac{1}{\sqrt{d_k}} \sum_k Q_{ik} K_{jk} \tag{17}$$

where $\sqrt{d_k}$ is the scaling factor, $Dot \in \mathbb{R}^{G\times 1\times(U+1)\times 2(U+1)}$, $Q \in \mathbb{R}^{G\times 1\times(U+1)\times C_s}$, and $K \in \mathbb{R}^{G\times 1\times 2(U+1)\times C_s}$. The attention scores are then calculated using a scaled Softmax function. Finally, the output of the attention mechanism is the weighted sum of the results, denoted as follows:

$$\begin{aligned} \text{Attn} &= \text{softmax}\left(\frac{\text{Dots}}{\sqrt{d_k}}\right) \\ \text{Out} &= \sum_{2(V+1)} \text{Attn} \end{aligned} \tag{18}$$

where the attention weights $Attn \in \mathbb{R}^{G\times 1\times(U+1)\times 2(U+1)}$ are used to perform a weighted sum on the values. The result of this weighted sum, $Out \in \mathbb{R}^{G\times 1\times(U+1)\times C_s}$, represents the aggregated context for the Q. These encodings help the model better understand the importance and contextual relationships of each position, allowing for more accurate attention score calculation. The scaling factor $(d_k)^{1/2}$ is used to scale the attention weights, ensuring that they are not influenced by the number of frequency bands.

Finally, PWSA establishes cross-group connections using group tokens. Specifically, in the CFAM stage, we obtain the output group tokens and then use these group tokens to compute the attention relationships between groups. After applying the GELU activation function and layer normalization, an attention weight matrix is generated. Attention is then computed between the attention weight matrix and the subfeature maps, facilitating the fusion of global information. The GELU activation function is described in the following equation:

$$\text{GELU} = x \cdot \frac{1}{2}\left[1 + \int_0^{x/\sqrt{2}} e^{-t^2}\,dt\right]. \tag{19}$$

GELU effectively combines the defined inactive information. LN stands for layer normalization, which improves the

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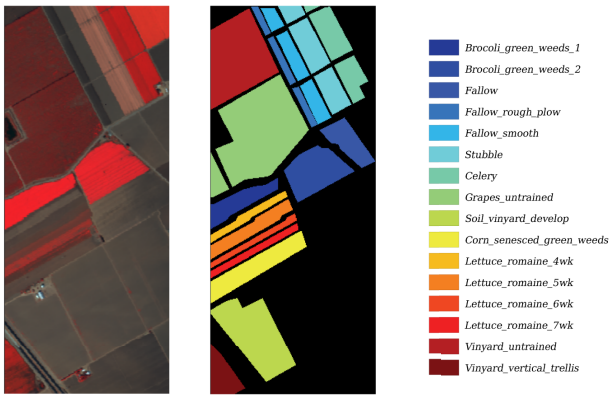


Fig. 4. (a) False-color composition of the AVIRIS Salinas scene. (b) Ground truth map containing 16 mutually exclusive land-cover classes.

stability and efficiency of model training by reparameterizing the distribution of the input data.

IV. Experiments

In this section, to validate the effectiveness of the proposed S2CIFTNet framework, we conducted comprehensive experiments on four public datasets and compared its performance with state-of-the-art methods. First, we provide a detailed introduction to the four datasets used in the experiments. Next, we describe the experimental setup in detail, including evaluation metrics, training details, and the selection of comparison methods. Then, we conduct an in-depth analysis of the experimental results to reveal the strengths and weaknesses of the framework. Finally, to further validate the contribution of each module in the framework, we designed a series of ablation experiments to assess the effectiveness and contribution of each module.

A. Four Publicly Available Datasets

1) Salinas: The SA dataset was collected in 1998 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in the western United States, California. The original SA dataset contains 224 bands with wavelengths ranging from 400 to 2500 nm. After excluding 20 water absorption bands, 204 spectral bands were retained for research and evaluation purposes. As shown in Fig. 4, this dataset consists of 512 $\times$ 217 pixels and includes 16 vegetation classes, with a total of 54 129 samples, distributed as follows: 1% for the training set, 1% for the validation set, and 98% for the test set.

2) Indian Pines: Indian Pines dataset was obtained in 1992 by NASA's AVIRIS instrument in the northwest region of Indiana. This dataset consists of a 145 $\times$ 145 pixel image, which includes 16 different land-cover classes. The spectral range spans from 0.4 to 2.5 μm, with 220 continuous spectral bands. Typically, 20 noisy bands are removed, leaving 200 spectral bands for research purposes. As shown in Fig. 5, a false-color composite image of the land-cover classes and the ground truth image of Indian Pines dataset are provided. This dataset contains 10 249 labeled samples, distributed as follows: 5% for the training set, 5% for the validation set, and 90% for the test set.

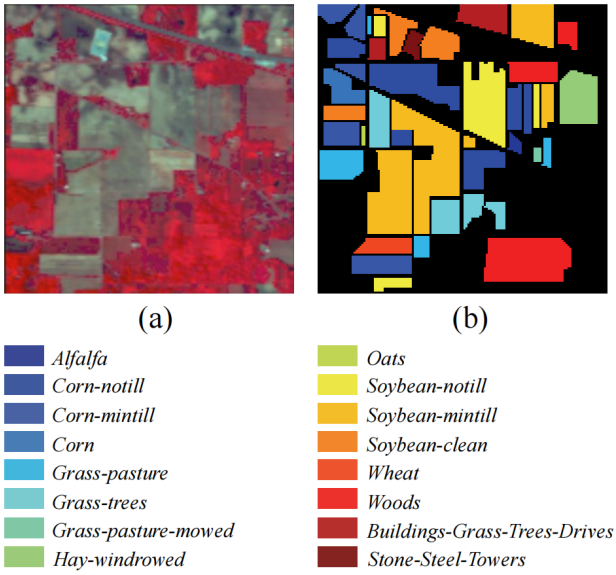


Fig. 5. (a) False-color composition of the AVIRIS Indian Pines scene. (b) Ground truth map containing 16 mutually exclusive land-cover classes.

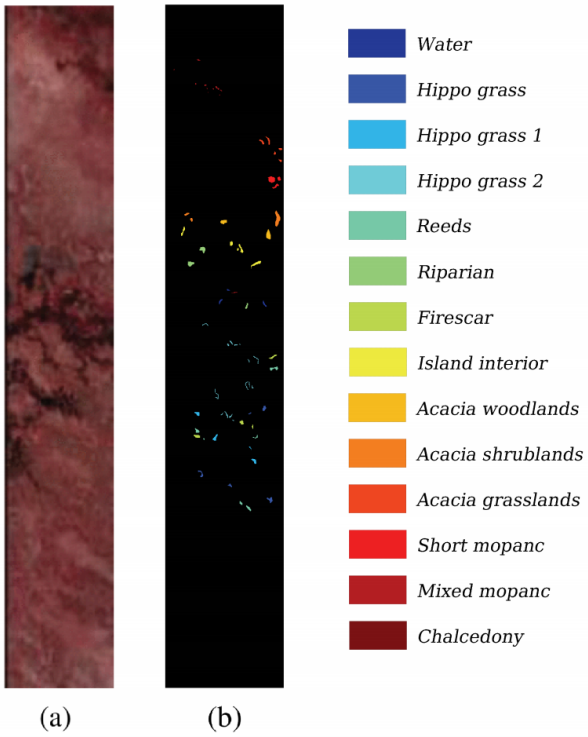


Fig. 6. (a) False-color composition of the AVIRIS Botswana scene. (b) Ground truth map containing 14 mutually exclusive land-cover classes.

3) Botswana: The Botswana dataset was obtained in May 2001 by NASA's Earth Observing-1 (EO-1) satellite, covering the Okavango Delta region in Botswana. The dataset has a size of 1476 $\times$ 256 pixels, with a wavelength range from 0.4 to 2.5 μm and a spatial resolution of approximately 20 m. Fig. 6 shows the pseudo-color composite image of the Botswana dataset along with the corresponding ground truth image. Out

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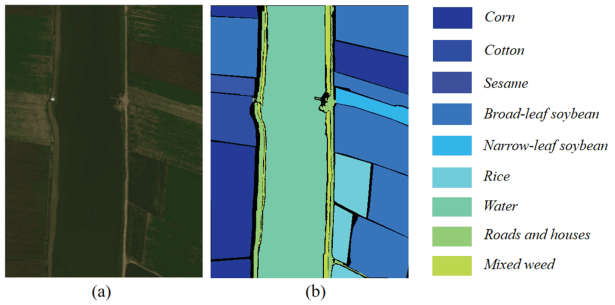


Fig. 7. (a) False-color composition of the AVIRIS WHU-Hi-LongKou scene. (b) Ground truth map containing nine mutually exclusive land-cover classes.

of the 242 spectral bands, five noisy bands were excluded: 1–9, 56–81, 98–101, 120–133, and 165–186, leaving 145 spectral bands. The study area includes 14 different land-cover types, including seasonal wetlands, intermittent wetlands, and dry forests at the distal end of the. The total number of samples is 3248. In the experimental setup, the dataset is divided into training 5%, validation 5%, and test 90% sets.

4) WHU-Hi-LongKou: The WHU-Hi-LongKou dataset was collected from a town in Hubei Province, China, using a hyperspectral imaging sensor mounted on the DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform. The dataset primarily includes six main crop classes: corn, cotton, sesame, broadleaf soybeans, narrowleaf soybeans, and rice. Fig. 7 shows the false-color composite image of the WHU-Hi-LongKou dataset along with the corresponding ground truth image. The collected image has a size of 550×400 pixels, with a spectral range from 0.4 to 1.0 μm, including 270 spectral bands, and a spatial resolution of approximately 0.463 m. The total number of samples is 204 542. In the experimental setup, the dataset is divided into the training set 0.2%, validation set 0.2%, and test set 99.6%.

B. Experimental Setup

1) Configuration: The previous version of the S2CIFTNet we proposed was developed using the PyTorch framework, with Python version 3.7.13 and PyTorch version 1.11. All experiments were conducted on a server equipped with a GeForce RTX 3090 GPU. For the S2CIFTNet method, we used the AdamW optimizer to update the network parameters, with the loss function being the cross-entropy loss. The learning rate was set to 0.001, the weight decay was set to 0.05, the batch size was set to 128, and the patch size was set to 8. To reduce the impact of randomness in the experiments and ensure the stability of the results, we repeated each experiment ten times and calculated the average of the results. Since different datasets have distinct characteristics, the feature extraction capability of the network may vary accordingly.

2) Evaluation Indicators: To strengthen the credibility of our method's performance, we used three comprehensive evaluation metrics: overall accuracy (OA), average accuracy (AA), and Kappa (k). The higher the values of these metrics, the better the classification performance of the model.

3) Comparison Methods: To highlight the current state of S2CIFTNet, we compare it with leading methods based on CNN and Transformer architectures as follows.

1) *2DCNN [15]:* The network model is mainly composed of 2-D convolutional layers and FC layers. After each convolution, there is a maximum pooling layer, and the final output is obtained through the FC layer.
2) *3DCNN [16]:* The network structure features a 3-D convolutional core, supplemented with 3-D padding layers and residual blocks. At the end, FC layers and a linear layer are connected for classification.
3) *HybridSN [17]:* There are three 3-D convolution blocks, one 2-D convolution block, and three FC layers.
4) *SSFTT [53]:* The network structure includes 3-D and 2-D convolutional layers for extracting low-level features, a Gaussian feature weighting layer for feature transformation, Transformer encoder layers, and a linear layer for classification.
5) *GAHT [54]:* The network is divided into three stages, each consisting of a 1×1 convolution embedded in grouped pixels and a Transformer encoder module. This is followed by a GAP layer and a linear classification layer.
6) *MHCFormer [51]:* The network combines hypergraph convolution with Transformer to achieve multimodal feature extraction and classification of hyperspectral images (HSI).
7) *MASSFormer [32]:* The framework concatenates memory tokens with raw features, feeding them into the memory-augmented Transformer encoder module for multilayer Transformer encoding. This process enhances the fusion and representation of spectral–spatial features through a memory-augmented mechanism.
8) *SSEFN [40]:* The network includes a spatial–spectral enhancement and fusion network, which enhances spectral spatial feature representation through spatial spectral enhancement strategies and utilizes adaptive decision dynamic fusion for multidecision prediction.
9) *S2VNet [36]:* The network includes a nonlinear autoencoder that extracts endmember and abundance information, combined with spectral variability parameter estimation and subpixel fusion module.
10) *MCTGCT [52]:* The network combines the advantages of CNNs and Transformers and combines graph contrastive learning.

C. Parameter Analysis

In order to gain a deeper understanding of how different parameter configurations affect the performance of the classification model, we carefully evaluate the classification performance of the S2CIFTNet model on four different datasets with varying numbers of attention heads and learning rate settings. Through this series of experiments, our goal is to reveal which parameter combinations can optimally improve the model's classification accuracy.

1) Number of Attention Heads: The number of attention heads is also crucial in our network classification. Our network

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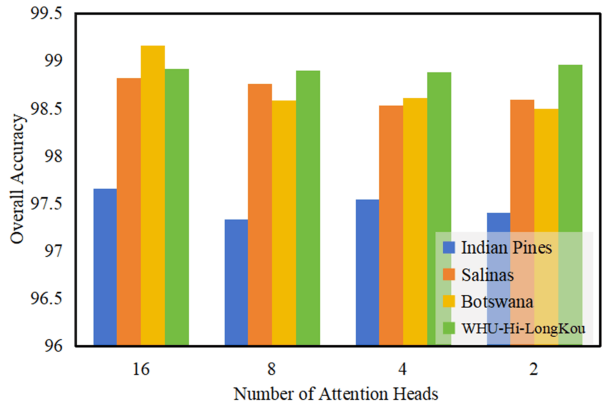


Fig. 8. Impact of four different numbers of attention heads on network performance on four datasets.

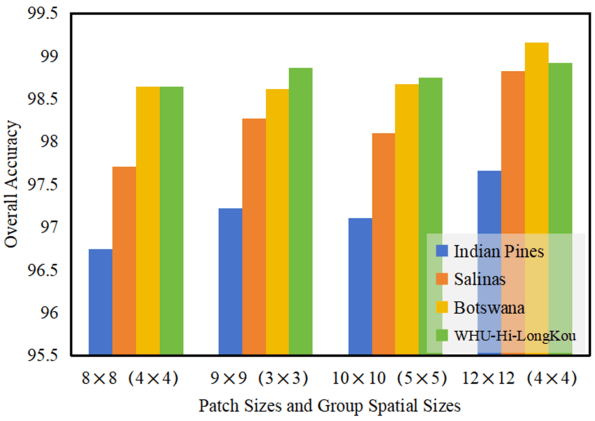


Fig. 9. Effect of four different patch sizes and group spatial sizes on network performance on four datasets.

has a hierarchical structure with decreasing channels at each layer, so the embedding dimensions of the IAF encoder for each stage are set to 128 and 64, respectively. The number of attention heads is implemented by channel splitting, and we follow the rule of gradually decreasing attention heads across stages, with the number of attention heads at each stage being half of the previous stage. We fixed other parameters and tested four different configurations for the number of attention heads, setting them to 16, 8, 4, and 2. As shown in Fig. 8, the best configuration for the number of attention heads across the four datasets is 16. Therefore, we set the number of attention heads to 16.

2) Patch Size and Group Spatial Sizes: To study the impact of patch size on classification performance, we tested four different patch sizes: 8, 9, 10, and 12, while keeping all other parameters constant. As shown in Fig. 9, the OA performance of these four patch sizes on the four datasets is provided. The results indicate that on the Botswana dataset, the OA value first decreases and then increases with the increase in patch size, reaching the best performance when the patch size is 12. On Indian Pine, Salinas, and WHU-Hi-LongKou datasets, the OA values show an alternating pattern of increase and decrease, with the highest value also achieved when the patch size is 12. Based on these results, we uniformly set the patch size to 12 for all four datasets.

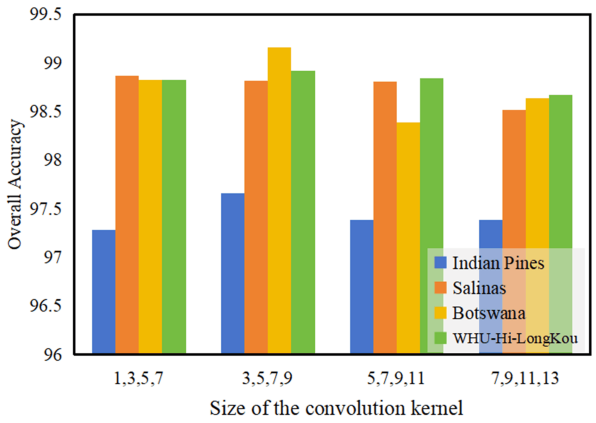


Fig. 10. Impact of four different convolution kernel sizes on network performance on four datasets.

3) Convolutional Kernel Size: In order to investigate the impact of convolutional kernel size on classification performance, we tested four configurations in the MS3 model, each comprising four convolutional kernels (k_1, k_2, k_3, and k_4) with sizes of (1, 3, 5, 7), (3, 5, 7, 9), (5, 7, 9, 11), and (7, 9, 11, 13), respectively, while strictly controlling all other parameters to remain constant. As shown in Fig. 10, the OA performance of these four convolution kernel sizes on four datasets is provided. The results show that on the Indian Pines and Botswana datasets, the OA value shows an alternating increase and decrease pattern, and the best performance is achieved when the size of the convolution core is $k_1 = 3$, $k_2 = 5$, $k_3 = 7$, and $k_4 = 9$. On the WHU-Hi-LongKou dataset, the OA value first increases, then decreases with the increase of convolution kernel size, and reaches its highest value when the convolution kernel size is $k_1 = 3$, $k_2 = 5$, $k_3 = 7$, and $k_4 = 9$. On the Salinas dataset, the OA value shows a continuous decreasing pattern with increasing kernel size. However, when the kernel size is $k_1 = 3$, $k_2 = 5$, $k_3 = 7$, and $k_4 = 9$, the OA value is only 0.05% less than when the kernel size is $k_1 = 1$, $k_2 = 3$, $k_3 = 5$, and $k_4 = 7$, so the kernel size is uniformly set to $k_1 = 3$, $k_2 = 5$, $k_3 = 7$, and $k_4 = 9$.

D. Classification Results and Analysis

To validate the effectiveness of this method, we selected ten representative network structures for comparative analysis. These network structures include 2DCNN, 3DCNN, and HybridSN CNN-based models, as well as Transformer-based models such as SSFTT, GAHT, MHCFormer, MASSFormer, SSEFN, S2VNet, and MCTGCL.

1) Analysis of Quantitative Results: In Tables V–VIII, we present the evaluation metrics of our method on four datasets—Indian Pine, Salinas, Botswana, and WHU-Hi-LongKou—including OA, AA, Kappa (k), and per-class accuracy. The best results are highlighted in bold.

As shown in Table V, experiments based on 5% training samples from the Indian Pine dataset demonstrate that our

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TABLE I

LAND-COVER CATEGORIES OF THE SALINAS DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Brocoli-green-weeds-1	20	20	1969
C02	Brocoli-green-weeds-2	37	37	6352
C03	Fallow	20	20	1936
C04	Fallow-rough-plow	14	14	1366
C05	Fallow-smooth	27	27	2624
C06	Stubble	40	39	3880
C07	Celery	36	36	3507
C08	Grapes-untrained	113	112	11046
C09	Soil-vinyard-develop	62	62	6079
C10	Corn-senesced-green-weeds	33	33	3212
C11	Lettuce-romaine-4wk	11	10	1047
C12	Lettuce-romaine-5wk	20	19	1888
C13	Lettuce-romaine-6wk	9	9	898
C14	Lettuce-romaine-7wk	10	11	1049
C15	Vinyard-untrained	73	72	7123
C16	Vinyard-vertical-trellis	18	18	1771
	Total	543	539	53047

TABLE II

LAND-COVER CATEGORIES OF INDIAN PINES DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Alfalfa	3	2	41
C02	Corn-notill	71	72	1285
C03	Corn-mintill	42	41	747
C04	Corn	12	12	213
C05	Grass-pasture	24	24	435
C06	Grass-trees	37	36	657
C07	Grass-pasture-mowed	2	1	25
C08	Hay-windrowed	24	24	430
C09	Oats	1	1	18
C10	Soybean-notill	49	48	875
C11	Soybean-mintill	122	123	2210
C12	Soybean-clean	30	29	534
C13	Wheat	10	10	185
C14	Woods	63	63	1139
C15	Buildings-Grass-Trees-Drives	19	20	347
C16	Stone-Steel-Towers	4	5	84
	Total	513	511	9225

TABLE III

LAND-COVER CATEGORIES OF THE BOTSWANA DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Water	14	13	243
C02	Hippo-grass	5	5	91
C03	Hippo grass-1	13	12	226
C04	Hippo grass-2	11	10	194
C05	Reeds	14	13	242
C06	Riparian	14	13	242
C07	Firescar	13	13	233
C08	Island interior	10	10	183
C09	Acacia woodlands	16	15	283
C10	Acacia shrulands	13	12	223
C11	Acacia grasslands	15	15	275
C12	Short mopanc	9	9	163
C13	Mixed mopanc	14	13	241
C14	Chalcedony	5	5	85
	Total	166	158	2924

TABLE IV

LAND-COVER CATEGORIES OF THE WHU-HI-LONGKOU DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CATEGORY

Class No.	Class name	Train	Valid	Test
C01	Corn	69	69	34373
C02	Cotton	17	16	8341
C03	Sesame	6	6	3019
C04	Broad-leaf soybean	127	126	62959
C05	Narrow-leaf soybean	9	8	4134
C06	Rice	24	23	11807
C07	Water	134	134	66788
C08	Roads and houses	15	14	7095
C09	Mixed weed	11	10	5208
	Total	412	406	203724

method achieves an OA of 97.66% $\pm$ 0.44%, improving by 0.8 percentage points compared to the next best method. When compared with CNN-based models like 2DCNN, 3DCNN, and HybridSN, our method overcomes the limitation of local receptive fields in capturing global information. For instance, HybridSN only achieves 44.35% $\pm$ 20.47% accuracy in classifying the first class, while our method achieves a much higher accuracy of 96.74% $\pm$ 2.96% in this category. In comparison with Transformer-based models, our method shows more stable cross-class performance. Models like MASSFormer excessively rely on spectral information for global modeling while neglecting spatial feature correlations, resulting in a significant error of 88.15% $\pm$ 5.11% in classifying the 12th class (vegetation area). Our method, through spectral–spatial joint encoding, improves the accuracy for this class to 94.86% $\pm$ 2.08%. Likewise, advanced models such as SSFTT, MHCFormer, SSEFN, and S2VNet, despite incorporating spatial attention mechanisms, fail to fully address multiscale features, thus affecting classification accuracy. Experimental data conclusively prove that our proposed dual-branch feature fusion architecture, which considers adaptive multiscale context aggregation and spectral–spatial attention collaborative optimization, has led to a groundbreaking improvement in HSIC performance.

In the Salinas dataset experiment, as indicated in Table VI, our method achieved an OA of 98.82% $\pm$ 0.32% using only 1% of the training samples, outperforming the second-best Transformer model by 0.32 percentage points. It achieved the highest accuracy in multiple classes, capturing complex dependencies between variables and providing rich statistical information. In the few-shot classification task, such as for class 8, it improved by 0.39 percentage points compared to the second-best model of the same class, demonstrating the strong ability of SODC to capture complex relationships between features. Although MCTGCL achieved a high accuracy of 100% $\pm$ 0.00% in class 6 by leveraging its expertise in modeling local features and several categories in SSEFN also attained high accuracy, its overall performance was limited due to the lack of multiscale feature recognition. Our method, on the other hand, employs the S2FDC module to achieve global context awareness, integrating spectral and depth convolutions with kernels of different sizes and orientations, effectively capturing multiscale spatial–spectral features. The OA of our method far exceeds that of other Transformer models, demonstrating significant advantages. Through the multiscale feature

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TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class	Convolution based Networks			Transformer based Networks							Our Methods
	2DCNN	3DCNN	HybridSN	SSFTT	GAHT	MHCFormer	MASSFormer	SSEFN	S2VNet	MCTGCL	Ours
1	66.52±22.30	56.96±8.90	44.35±20.47	65.12±20.17	63.90±14.95	78.48±22.07	68.54±18.78	87.80±4.8	63.91±20.35	73.64 ± 20.83	**96.74±2.96**
2	86.38±1.59	71.35±4.67	84.75±2.87	91.35±2.12	92.42±1.91	96.84±1.16	92.06±2.21	79.83±8.7	93.43±2.35	95.65 ± 1.18	**97.65±0.84**
3	90.86±4.77	71.39±7.80	79.19±6.98	93.07±4.58	95.48±2.30	96.75±1.42	96.49±1.50	68.13±7.2	95.25±1.78	**98.06 ± 1.35**	97.66±1.77
4	92.11±3.78	25.65±7.10	68.31±7.68	91.88±3.20	95.45±3.37	95.36±3.58	93.29±5.04	81.29±7.7	93.71±4.14	94.78 ± 2.51	**97.26±3.01**
5	92.13±3.38	87.25±13.95	89.50±4.06	92.71±3.44	93.91±3.50	96.05±2.31	89.47±4.61	87.57±7.6	92.69±3.25	**98.32 ± 1.17**	96.40±2.53
6	96.97±1.89	95.16±2.56	96.73±2.15	98.33±1.10	98.61±0.98	**99.55±0.36**	97.69±1.25	98.33±1.8	97.84±1.00	97.97 ± 0.48	98.84±1.37
7	**98.21±4.30**	76.79±12.11	82.50±21.46	84.80±14.73	84.80±11.70	69.29±25.71	81.60±13.76	37.50±24.73	81.07±21.07	84.44 ± 26.90	86.07±11.79
8	96.99±2.98	97.01±1.53	98.37±1.11	99.30±1.04	99.74±0.51	**100.00±0.00**	99.00±1.29	99.78±1.67	98.83±1.53	99.80 ± 0.32	99.81±0.27
9	52.50±25.02	40.00±6.07	53.50±21.57	56.11±20.71	30.00±8.68	38.50±22.14	59.44±29.09	29.41±27.65	56.00±13.93	59.47 ± 24.35	**63.00±21.00**
10	86.21±4.80	73.16±2.92	82.98±5.29	88.03±3.97	90.97±5.04	92.98±33.78	91.35±3.91	48.86±25.84	89.86±4.32	**96.70 ± 4.06**	95.91±2.71
11	93.30±1.79	84.74±4.20	89.34±2.35	95.03±1.60	96.49±1.87	97.82±0.86	96.87±1.26	80.24±10.9	96.20±0.97	97.78 ± 0.95	**98.22±0.78**
12	83.44±3.52	70.19±5.99	75.68±7.85	88.46±4.49	92.85±3.60	**95.35±2.12**	88.15±5.11	54.70±29.7	94.22±2.59	92.70 ± 1.73	94.86±2.08
13	98.39±1.29	95.02±9.56	98.78±0.91	**99.41±1.09**	99.14±1.16	99.32±0.79	98.32±1.85	94.66±1.94	98.59±2.50	98.77 ± 1.34	98.24±2.03
14	96.35±1.55	93.68±3.86	96.17±2.54	97.41±1.73	98.13±0.67	**99.49±0.47**	97.34±1.48	95.21±4.52	98.12±0.96	99.43 ± 0.32	99.29±0.78
15	87.41±7.12	88.06±8.35	76.61±5.40	87.09±5.85	90.40±4.82	93.16±4.38	87.44±8.34	70.90±16.39	90.03±4.63	93.24 ± 2.69	**96.97±3.16**
16	**96.77±2.10**	90.22±5.94	94.52±8.32	94.29±4.90	93.57±6.70	87.10±10.11	85.12±8.80	69.16±26.67	95.59±4.15	82.13 ± 5.58	93.01±6.55
OA(%)	91.35±0.73	83.59±1.01	87.26±1.49	93.41±0.65	94.96±0.76	96.76±0.48	94.25±0.47	81.01±1.58	94.82±0.44	96.86 ± 0.98	**97.66±0.44**
AA(%)	88.41±1.61	74.50±1.55	81.96±3.33	88.90±2.17	88.49±1.16	89.75±2.72	88.89±1.27	77.96±2.93	89.71±2.43	91.43 ± 4.12	**94.53±1.52**
$k\times$ 100	90.13±0.83	80.02±1.17	85.44±1.72	92.48±0.74	94.24±0.87	96.31±0.55	93.45±0.54	78.97±1.91	94.09±0.50	96.42 ± 1.12	**97.33±0.50**

TABLE VI

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE SALINAS DATASET

Class	Convolution based Networks			Transformer based Networks							Our Methods
	2DCNN	3DCNN	HybridSN	SSFTT	GAHT	MHCFormer	MASSFormer	SSEFN	S2VNet	MCTGCL	Ours
1	53.99±45.71	97.71±1.32	99.37±0.71	99.77±0.38	99.40±1.21	98.85±2.23	96.72±4.11	**100.00 ± 0.00**	99.78±0.32	99.38 ± 0.25	99.83±0.52
2	82.65±32.33	99.88±0.09	99.74±0.20	99.82±0.22	99.79±0.32	99.76±0.39	99.39±1.01	99.09±0.33	99.89±0.20	99.99 ± 0.01	**100.00±0.00**
3	83.29±27.88	89.17±3.37	96.76±3.25	98.89±0.58	98.93±1.17	96.99±2.27	96.94±1.55	98.79±0.39	97.77±1.83	97.98 ± 0.03	**99.55±0.83**
4	97.39±5.48	98.80±1.13	97.84±3.18	99.00±1.34	98.67±2.52	96.32±1.35	97.00±2.87	96.97±2.39	98.60±1.86	**99.76 ± 0.16**	99.26±1.11
5	84.48±23.00	92.61±3.14	94.33±2.90	98.01±2.11	97.60±2.11	96.74±3.69	97.03±2.12	98.06±0.83	96.53±2.72	**99.65 ± 0.22**	98.21±1.92
6	99.97±0.07	99.96±0.10	99.75±0.47	99.84±0.23	99.88±0.36	99.93±0.22	99.76±0.49	**100.00 ± 0.00**	99.89±0.22	**100.00 ± 0.00**	99.94±0.12
7	98.04±4.32	99.73±0.18	99.23±0.57	99.88±0.08	99.90±0.16	99.61±0.82	99.25±0.88	**100.00 ± 0.00**	99.73±0.34	98.79 ± 0.51	99.56±1.18
8	85.36±7.72	85.43±2.24	86.04±3.33	91.45±1.71	92.68±3.19	92.32±2.68	93.29±2.36	76.14±3.67	92.51±2.11	97.38 ± 0.75	**97.77±1.23**
9	97.34±3.87	98.64±1.00	99.37±0.43	99.69±0.43	99.87±0.20	99.81±0.23	99.68±0.36	99.53±0.39	99.70±0.50	**100.00 ± 0.00**	99.98±0.03
10	89.38±7.87	90.56±4.31	92.48±2.43	96.44±1.11	98.49±0.87	96.35±2.14	97.57±1.37	96.38±2.34	96.65±1.40	97.34 ± 0.76	**98.54±1.90**
11	50.86±34.71	72.13±8.43	90.10±9.11	93.32±7.24	95.82±5.32	93.26±4.45	94.52±4.47	94.04±4.44	92.31±8.64	**99.33 ± 0.47**	99.26±1.06
12	83.43±19.91	98.41±0.71	99.35±0.93	99.78±0.24	**99.96±0.05**	99.90±0.16	99.34±1.46	99.94±0.11	99.33±0.99	99.10 ± 0.18	99.78±0.31
13	89.18±29.76	99.89±0.26	98.14±1.69	99.31±1.45	99.50±1.00	99.28±0.55	99.55±0.57	71.11±4.62	99.00±1.33	96.59 ± 1.39	**99.90±0.26**
14	92.90±9.82	99.02±0.61	97.18±1.19	97.78±2.35	**99.39±0.63**	99.08±0.51	97.66±2.23	83.97±8.87	98.99±0.70	98.99 ± 0.48	99.27±0.95
15	67.62±25.55	73.87±4.49	81.46±3.63	83.85±4.12	92.39±2.83	88.89±5.61	92.57±2.11	57.44±6.80	88.66±3.81	**98.50 ± 0.46**	97.03±1.78
16	80.69±13.86	90.35±3.95	96.41±2.22	97.01±1.64	98.18±1.16	97.14±2.81	96.05±3.89	**100.00 ± 0.00**	97.07±2.18	99.14 ± 0.12	99.23±0.87
OA(%)	84.65±8.33	90.87±0.66	93.09±0.54	95.30±0.35	96.94±0.53	96.02±0.56	96.54±0.72	87.26±1.25	96.04±0.66	98.50 ± 0.41	**98.82±0.32**
AA(%)	83.53±10.72	92.88±0.88	95.47±0.68	97.12±0.36	98.15±0.32	97.33±0.47	97.27±0.79	91.96±0.99	97.28±0.56	98.76 ± 0.32	**99.19±0.16**
$k\times$ 100	82.84±9.37	89.83±0.73	92.30±0.59	94.76±0.39	96.59±0.59	95.57±0.63	96.15±0.80	85.84±0.85	95.59±0.74	98.15 ± 0.42	**98.68±0.35**

TABLE VII

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE BOTSWANA DATASET

Class	Convolution based Networks			Transformer based Networks							Our Methods
	2DCNN	3DCNN	HybridSN	SSFTT	GAHT	MHCFormer	MASSFormer	SSEFN	S2VNet	MCTGCL	Ours
1	99.85±0.34	**100.00±0.00**	**100.00±0.00**	99.79±0.42	99.34±0.59	99.85±0.25	98.19±1.38	**100.00 ± 0.00**	99.81±0.45	**100.00 ± 0.00**	98.78±1.44
2	86.14±7.88	88.42±8.08	96.53±4.16	97.80±1.97	**100.00±0.00**	86.93±18.22	93.52±7.65	**100.00 ± 0.00**	97.23±5.10	**100.00 ± 0.00**	99.80±0.59
3	87.37±5.52	97.05±2.85	98.96±1.22	99.82±0.53	**100.00±0.00**	99.48±1.43	97.04±3.00	**100.00 ± 0.00**	99.92±0.24	95.98 ± 5.96	**100.00±0.00**
4	86.56±4.17	97.35±1.52	98.05±1.57	99.79±0.47	99.59±0.94	99.77±0.56	97.78±4.10	97.15±0.63	99.67±0.55	**100.00 ± 0.00**	99.91±0.28
5	88.22±6.32	85.99±5.98	88.40±2.56	95.12±4.61	95.45±4.48	94.20±5.72	89.38±9.50	81.29±2.71	95.13±3.18	**99.73 ± 0.46**	96.84±2.17
6	81.90±9.35	73.57±6.48	90.74±5.12	96.98±2.08	97.19±2.06	95.58±6.91	90.45±4.32	89.47±3.68	96.02±2.42	96.68 ± 1.31	**98.36±1.50**
7	95.17±4.03	99.50±0.49	99.85±0.26	99.91±0.17	**100.00±0.00**	**100.00±0.00**	98.93±1.60	**100.00 ± 0.00**	99.81±0.46	**100.00 ± 0.00**	99.81±0.58
8	77.64±8.97	86.31±9.69	97.98±4.45	96.89±6.66	99.40±1.63	96.80±8.38	95.30±9.19	**100.00 ± 0.00**	98.37±4.56	**100.00 ± 0.00**	99.85±0.44
9	79.14±6.91	95.57±4.36	93.22±3.24	97.63±2.54	99.43±0.85	99.14±1.02	98.48±1.47	75.77±3.19	98.57±1.72	**100.00 ± 0.00**	99.71±0.39
10	96.45±2.33	96.81±1.53	96.33±3.90	99.19±0.69	**100.00±0.00**	99.56±0.87	99.10±1.32	**100.00 ± 0.00**	99.35±1.23	**100.00 ± 0.00**	99.92±0.24
11	97.90±1.05	99.05±2.04	99.38±1.05	97.82±1.96	99.24±0.55	99.31±1.04	96.65±3.08	**100.00 ± 0.00**	99.64±0.77	99.90 ± 0.31	99.41±1.14
12	95.80±2.39	99.34±0.48	97.96±2.87	**100.00±0.00**	99.88±0.25	99.06±2.64	94.79±3.49	95.53±2.72	99.61±0.99	69.36 ± 1.16	99.06±1.91
13	96.75±1.38	98.92±1.34	97.72±3.09	99.71±0.87	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00 ± 0.00**	**100.00±0.00**
14	88.53±6.11	87.68±5.22	93.37±3.65	94.82±6.03	90.35±6.94	89.76±7.72	94.00±5.88	**100.00 ± 0.00**	94.00±5.88	98.68 ± 1.83	94.42±6.16
OA(%)	90.11±2.11	93.74±0.73	96.33±0.80	98.36±0.72	98.86±0.40	97.93±1.08	96.10±1.47	93.98±1.23	98.60±0.61	97.64 ± 0.42	**99.16±0.41**
AA(%)	89.82±1.90	93.25±0.82	96.32±0.91	98.24±0.72	98.56±0.57	97.07±1.57	95.67±1.76	95.65±1.93	98.37±0.75	97.17 ± 0.40	**98.99±0.62**
$k\times$ 100	89.28±2.28	93.22±0.80	96.02±0.87	98.23±0.78	98.76±0.43	97.76±1.17	95.78±1.59	93.47±1.44	98.49±0.67	97.44 ± 0.46	**99.09±0.44**

fusion mechanism enhanced by spatial–spectral augmentation, we achieve fine-grained modeling of spatial–spectral features, effectively overcoming the limitations of traditional methods in global context awareness.

In the experiment on the Botswana dataset, as shown in Table VII, our method achieved an OA of 99.16% ± 0.41% using only 5% of the training samples, demonstrating more robust cross-class generalization performance. Due to the discrete nature of feature distribution and the irregular shapes of classes, CNN-based methods face challenges in modeling global context information and show insufficient sensitivity to recognizing targets at different scales. Therefore, the method proposed in this article optimizes the feature extraction and fusion mechanisms, achieving the best single-class

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TABLE VIII
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE WHU-HI-LONGKOU DATASET

Class	Convolution based Networks			Transformer based Networks							Our Methods
	2DCNN	3DCNN	HybridSN	SSFTT	GAHT	MHCFormer	MASSFormer	SSEFN	S2VNet	MCTGCL	Ours
1	97.68±2.31	98.45±0.84	98.81±0.86	99.54±0.26	99.77±0.16	99.63±0.34	99.37±0.36	99.61±0.27	99.55±0.21	**99.94 ± 0.05**	99.80±0.10
2	74.35±29.55	83.07±7.41	89.10±4.18	97.83±1.16	97.10±1.72	98.64±0.71	93.85±3.11	96.68±2.86	95.00±3.46	98.55 ± 0.89	**99.15±0.58**
3	3.41±10.22	60.97±9.67	83.32±3.24	89.96±5.73	91.18±7.24	91.73±3.66	87.48±4.87	86.96±5.31	90.19±6.20	**97.95 ± 1.38**	94.18±4.37
4	94.82±2.90	97.33±1.14	97.23±0.90	98.77±0.58	98.52±0.78	99.38±0.45	98.17±0.54	98.11±0.91	98.90±0.49	**99.92 ± 0.04**	99.31±0.40
5	2.15±4.43	35.48±20.78	72.78±8.16	89.20±4.96	90.52±4.78	88.13±6.11	83.41±6.88	45.17±9.95	86.89±4.82	88.41 ± 5.70	**93.12±4.26**
6	77.98±28.25	97.34±2.49	98.18±1.78	98.55±1.45	99.06±1.34	98.86±0.95	97.47±2.34	**99.62±0.79**	98.57±1.40	98.83 ± 0.37	98.20±1.81
7	99.90±0.15	99.97±0.03	**99.98±0.02**	99.90±0.11	99.77±0.25	99.93±0.06	99.55±0.36	99.12±0.13	99.89±0.19	99.90 ± 0.08	99.92±0.07
8	82.14±13.42	85.96±3.92	91.23±3.00	90.73±5.28	89.50±4.53	93.48±2.93	82.35±6.67	91.99±3.88	93.78±2.19	84.10 ± 1.97	**94.12±3.18**
9	43.36±24.27	70.61±9.14	79.84±6.40	89.88±3.40	91.23±2.91	91.45±3.94	84.37±5.70	**93.67±1.44**	86.30±5.20	88.84 ± 2.69	90.52±3.99
OA(%)	90.16±4.03	94.93±0.69	96.76±0.34	98.39±0.30	98.34±0.45	98.79±0.18	97.25±0.40	95.98±0.78	98.28±0.33	98.70 ± 0.11	**98.92±0.23**
AA(%)	63.98±9.18	81.02±3.27	90.05±0.88	94.93±1.33	95.18±1.60	95.69±0.88	91.78±1.62	90.10±1.87	94.34±1.24	95.16 ± 0.60	**96.48±1.08**
$k\times$ 100	86.91±5.44	93.30±0.91	95.74±0.44	97.88±0.40	97.82±0.59	98.41±0.24	96.38±0.52	94.75±0.74	97.74±0.44	98.29 ± 0.14	**98.58±0.30**

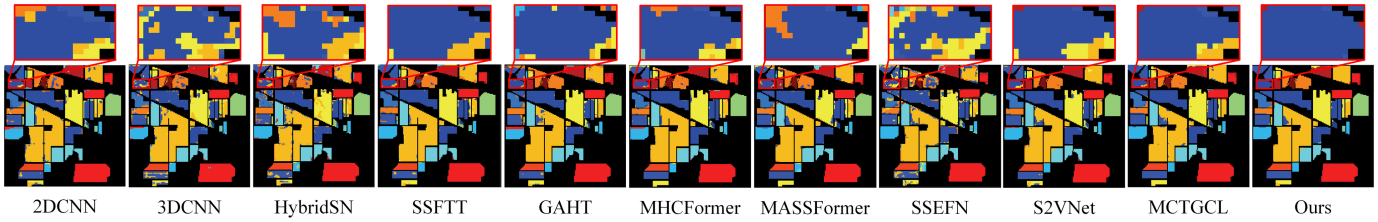


Fig. 11. Indian Pines dataset visualization classification map.

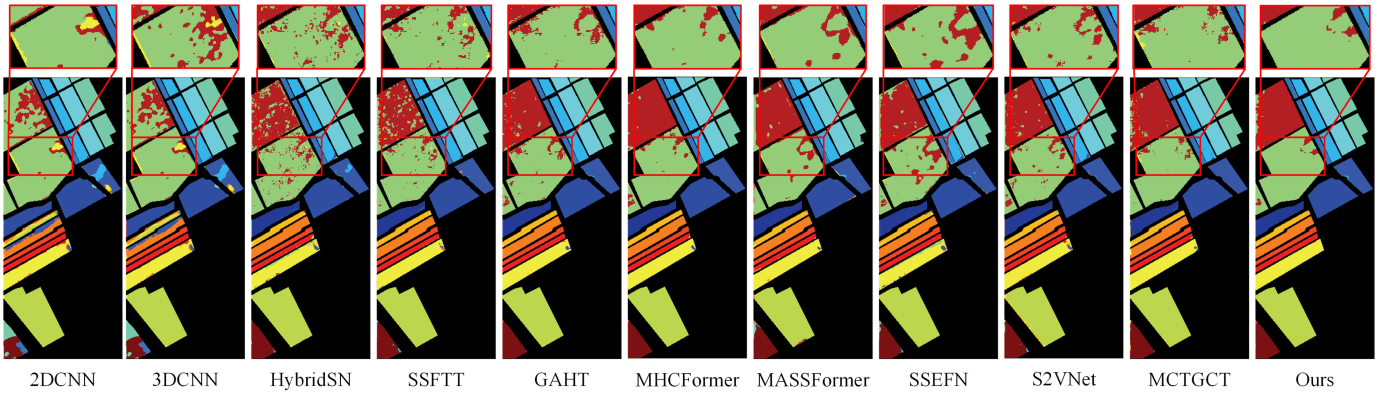


Fig. 12. Salinas dataset visualization classification map.

classification accuracy across multiple classes. Although GAHT achieved perfect accuracy of 100.00% ± 0.00% in several categories, compared to our method, it fails to effectively recognize features at different scales, resulting in lower OA, AA, and Kappa coefficient.

In the WHU-Hi-LongKou dataset, as shown in Table VIII, our method achieved the highest accuracy in nearly all categories. The results indicate that, for the fifth class of narrow-leafed soybeans, although it shares high similarity with the fourth class, our method, despite having very few samples, achieved the best single-class accuracy. This can be attributed to the significant advantage of our method in recognizing feature classes at different scales. Additionally, the S2FDC module in our method helps capture subtle spectral variations, enabling more effective differentiation of features. In this dataset, we used only 0.2% of the training samples. Despite the limited sample size, the OA reached 98.92% ± 0.23%, which is approximately 0.13% higher than the second-best method, MHCFormer. At the same time, our method also achieved an AA of 96.48% ± 1.08%, demonstrating excellent average performance.

2) Visualization and Analysis of Results: In order to more clearly demonstrate the superiority of the proposed method compared to other methods, Figs. 11–14 show the classification visualization results of all methods across four datasets. It is evident that the classification maps generated by our method, S2CIFTNet, contain significantly less noise and are closer to the true ground conditions. Taking the Salinas dataset as an example, Fig. 12 clearly shows that, for enlarged illustrations of the eighth class of Grapes_untrained, the classification map produced by our method is exceptionally clear both on the surface and at the edges, with the least amount of noise. In contrast, the other ten comparative methods exhibit varying degrees of noise on the surface and edges of this class of objects. At the same time, the classification map generated by our method has clear boundary lines, making it easier for the model to recognize real edge pixels, and our recall rate is also higher than that of other methods. Consequently, the

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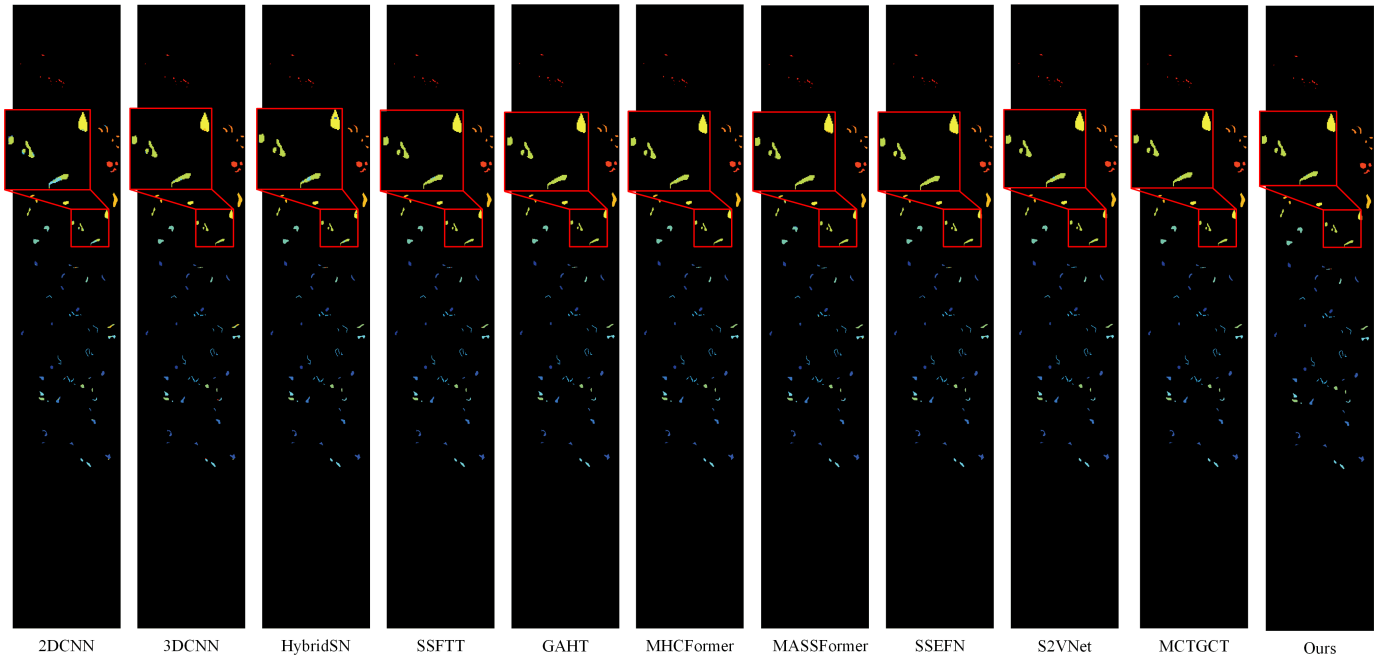


Fig. 13. Botswana dataset visualization classification map.

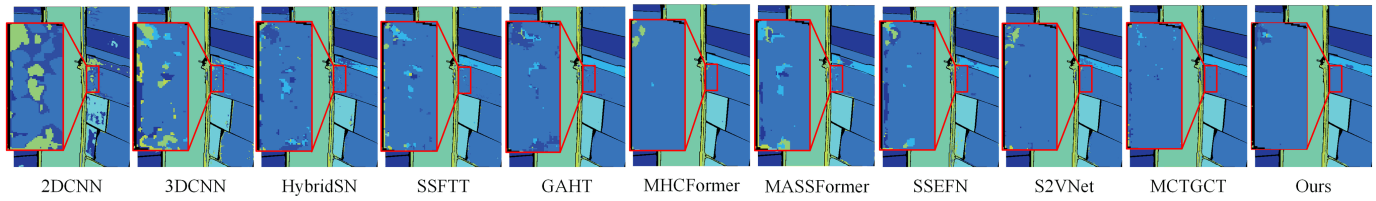


Fig. 14. WHU-Hi-LongKou dataset visualization classification map.

TABLE IX
PERFORMANCE COMPARISON OF 11 MODELS ON FOUR HSI DATASETS

Methods	Indian Pines			Salinas			Botswana			WHU-Hi-LongKou		
	Parameters/K	FLOPs/MMac	Latency/ms	Parameters/K	FLOPs/MMac	Latency/ms	Parameters/K	FLOPs/MMac	Latency/ms	Parameters/K	FLOPs/MMac	Latency/ms
2DCNN	1708.944	66.441	69.03	1718.160	67.556	69.79	1581.966	51.108	49.30	1869.321	85.955	85.23
3DCNN	**161.676**	79.484	77.87	**263.596**	138.247	126.18	**179.594**	98.371	93.38	**201.989**	182.711	162.57
HybridSN	512.220	247.682	214.21	512.220	247.682	214.21	512.220	247.682	214.21	512.220	247.682	214.21
SSFTT	931.848	49.382	54.11	950.280	50.374	54.95	678.278	35.743	42.62	1253.953	66.738	68.73
GAHT	830.8	159.35	144.16	972.62	62.18	64.65	800.72	51.2	55.44	1514.2	96.85	93.73
MHCFormer	3804.720	312.745	273.52	3892.608	318.236	278.12	2804.990	227.779	178.41	5108.665	419.425	362.56
MASSFormer	304.5	**25.54**	**47.33**	309.86	**25.54**	47.33	308.58	**25.54**	47.33	314.37	**25.54**	47.34
SSEFN	1996.291	102.8	99.19	1996.291	102.8	99.19	1995.036	102.8	99.19	1991.899	102.8	99.19
S2VNet	604.5	61.64	63.77	604.5	61.64	63.77	508.58	61.64	63.66	514.37	61.64	63.66
MCTGCL	271.44	33.28	56.15	271.44	33.28	56.15	272.18	33.28	56.15	271.54	33.28	56.15
Ours	539.966	60.628	53.86	368.99	51.318	**46.02**	367.836	51.170	**45.90**	407.799	55.153	**47.21**

boundary $F1$-score reaches 98.9% for S2CIFTNet, surpassing the second-ranked MCTGCL method by 0.4%. The same observation holds true for the other three datasets as well. This demonstrates that our method not only outperforms others in classification accuracy but also produces cleaner and more precise classification maps, especially in areas where noise could otherwise affect the results.

3) Calculation Cost Analysis: We evaluated the complexity of ten methods on four datasets, including floating-point operations (FLOPs) and the number of parameters (Params), as shown in Table IX. The results indicate that on the Salinas, Botswana, and WHU-Hi-LongKou datasets, our model requires fewer parameters and FLOPs compared to other models, suggesting that we achieved higher classification accuracy with a model that has a lower parameter count and computational cost. Our method has more parameters compared to 3DCNN, MASSFormer, and MCTGCL, because their architectures are relatively simple, which prevents them from achieving high accuracy across all categories. For example, MASSFormer, which has the fewest FLOPs, has an accuracy of 96.54% $\pm$ 0.72% on the Salinas dataset. Although our FLOPs are slightly higher than theirs, our accuracy on the Salina dataset reaches 98.82% $\pm$ 0.32%, with a 0.07 increase in accuracy for each additional floating-point number. MHCFormer, which has the most parameters and FLOPs, has a much lower accuracy than us in the four datasets. Notably,

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TABLE X
ANALYSIS OF ABLATION OF DIFFERENT COMPONENTS IN THE MODEL

Methods	S2FDC Module	IAF Module	Evaluation Indicators		
			OA(%)	AA(%)	Kappa(k)
Indian Pines	×	×	96.30±0.43	91.68±1.80	95.78±0.50
	✓	×	96.42±0.94	92.07±1.96	95.91±1.07
	×	✓	97.34±0.51	93.38±2.40	96.97±0.58
	✓	✓	**97.66±0.44**	**94.53±1.52**	**97.33±0.50**
Salinas	×	×	96.40±0.37	97.66±0.69	96.00±0.41
	✓	×	98.28±0.38	98.71±0.20	98.09±0.42
	×	✓	98.80±0.39	99.16±0.23	98.66±0.43
	✓	✓	**98.82±0.32**	**99.19±0.16**	**98.68±0.35**
Botswana	×	×	97.90±0.83	97.76±0.92	97.72±0.90
	✓	×	98.48±0.43	98.23±0.56	98.35±0.47
	×	✓	98.92±0.49	98.75±0.67	98.83±0.53
	✓	✓	**99.16±0.41**	**98.99±0.62**	**99.09±0.44**
WHU-Hi-LongKou	×	×	98.47±0.36	94.73±1.53	97.98±0.47
	✓	×	98.51±0.27	95.45±1.20	98.04±0.35
	×	✓	98.86±0.23	96.31±0.81	98.50±0.30
	✓	✓	**98.92±0.23**	**96.48±1.08**	**98.58±0.30**

after unifying experimental configurations (e.g., batch sizes), our method achieves state-of-the-art inference latency across three datasets. While its latency on Indian Pines dataset is marginally higher than MASSFormer, our framework demonstrates dual advantages in edge deployment scenarios with constrained computational resources. Run faster than other models while maintaining leading classification accuracy. These characteristics position our method as a preferred solution for real-time hyperspectral applications requiring both precision and computational efficiency. It is important to emphasize that, although S2CIFTNet may not be the most efficient, it achieves state-of-the-art accuracy on all datasets with relatively fewer inference latency.

4) Robustness Analysis: To validate the effectiveness of the two main components of our proposed method, we conducted ablation experiments on four datasets to assess the impact of each component on classification performance. Our model consists of two main components: the S2FDC module and the IAF module. The evaluation metrics include OA, AA, and Kappa. The experimental results are shown in Table X, and an analysis is provided as follows, using the Salinas dataset as an example. The first row shows the results without any modules; the second row shows the results without the IAF module, the third row shows the results without the S2FDC module, and the fourth row shows the results with both the S2FDC and IAF modules.

The results of the study indicate that the metrics in the first row, which represent the lowest values, are inferior compared to the second and third rows, where the results show significant improvement. This demonstrates the substantial impact of the modules we designed. Furthermore, the results of the second and third rows show that using the single IAF module outperforms using only the S2FDC module. By combining the data from the second, third, and fourth rows, we find that the feature extraction capability of relying on a single module is inferior to the combined effect of both modules, highlighting the importance of the S2FDC and IAF modules in handling feature relationships at different scales. Ultimately, the data clearly shows that when both modules are present, the OA improves by 0.54% and 0.02% compared to the cases where only the S2FDC or IAF module is added, and improves by 2.42% compared to the case with no modules added. This result once again significantly demonstrates the effectiveness of the two main components in our method (this holds true for the other three datasets as well).

TABLE XI
ABLATION ANALYSIS OF SUBMODULES IN THE MODEL

Methods	SODC Block	MS3 Block	HSAC Block	CFAM Block	OA(%)
Indian Pines	×	✓	✓	✓	97.22±0.37
	✓	×	✓	✓	97.16±0.48
	✓	✓	×	✓	97.17±0.53
	✓	✓	✓	×	97.49±0.48
	✓	✓	✓	✓	**97.66±0.44**
Salinas	×	✓	✓	✓	98.79±0.31
	✓	×	✓	✓	98.79±0.34
	✓	✓	×	✓	98.72±0.35
	✓	✓	✓	×	98.80±0.42
	✓	✓	✓	✓	**98.82±0.32**
Botswana	×	✓	✓	✓	98.62±0.48
	✓	×	✓	✓	98.65±0.65
	✓	✓	×	✓	98.66±0.49
	✓	✓	✓	×	98.95±0.59
	✓	✓	✓	✓	**99.16±0.41**
WHU-Hi-LongKou	×	✓	✓	✓	98.85±0.30
	✓	×	✓	✓	98.77±0.24
	✓	✓	×	✓	98.88±0.17
	✓	✓	✓	×	98.91±0.26
	✓	✓	✓	✓	**98.92±0.23**

The two main parts of our module, S2FDC module and IAF module, have their own separable submodules, namely, SODC block and MS3 block of S2FDC module, and HSAC block and CFAM block of IAF module. The evaluation metric is OA. The experimental results are shown in Table XI. Taking Indian Pines dataset as an example, the first row shows the results without SODC block; the second line shows the result without MS3 block. The third line shows the result without HSAC block; The fourth line shows the result without CFAM block; The fifth line displays the results that are available for all blocks.

The research results show that the indicators in the first four rows are not as good as those in the fifth row, indicating the substantial impact of the submodule we designed. The data clearly show that when all four modules are present, the OA improves by 0.44%, 0.50%, 0.49%, and 0.17%, respectively, compared to the case where SODC, MS3, HSAC, and CFAM models are not added, and by 1.36% compared to the case where no model is added. This result once again significantly demonstrates the effectiveness of the submodules in our method (the same applies to the other three datasets).

V. CONCLUSION

In this study, we propose the S2CIFTNet model, which consists of a multiscale spatial–spectral information interaction module, an SODC block, two feature extraction blocks, a GAP block, and an FC layer. The model combines CNN and Transformer to deeply mine the multiscale spatial and spectral information in hyperspectral images (HSI) and reveals the correlation between spectral channels through covariance analysis, thereby providing the model with a deeper feature

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understanding. In the multiscale spatial–spectral information interaction module, we extract multiscale features of HSI from different perspectives and use the second-order distance CM to explore the correlation between spectral channels, enhancing the model's sensitivity to spectral information, which lays the foundation for subsequent feature processing. The two feature extraction blocks mainly consist of the HSAC block and NRL. The HSAC block includes Pconv, 2Dconv, and DWconv, which effectively enhance the extraction of multiscale spatial features. The NRL block combines 2Dconv, average pooling, and max pooling to optimize the model's performance, effectively retaining key features and improving the model's discriminative ability. We conducted extensive experiments on four datasets, and the results show that the model can maintain high classification accuracy with lower parameters and computational cost. In the future, we will continue to explore the advantages of CNN and Transformer to better apply multiscale representations to HSIC tasks.

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